
September 9, 2026

Fractal Dimension

Let's talk about a property of fractals. . .



Fractal Dimension

What does it mean for a shape to have a **non-integer** dimension?

Dimensions We Know

We are all familiar with integer dimensions:



Dimensions We Know

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Line



1D



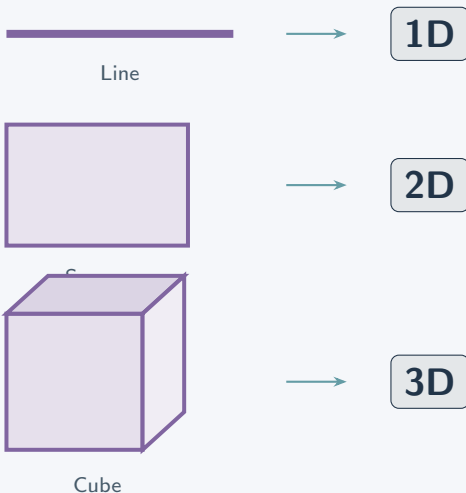
Square



2D

Dimensions We Know

We are all familiar with integer dimensions:



Dimensions are always whole numbers. . . right?

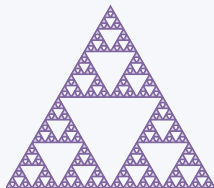
What About This One?

What dimension does this shape
have?

Any guesses?

- It lives in 2D space. . .
- But it has **zero area**
- And **infinite perimeter**

It's not quite 2D, but it's clearly more than
1D. . .



Sierpinski Triangle

What About This One?

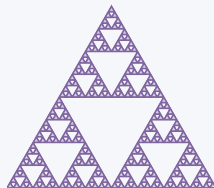
Sierpinski Dimension

The Sierpinski Triangle has a fractal dimension of:

$$D \approx 1.585$$

!!

A **non-integer** dimension — between a line (1D) and a filled surface (2D).



$$D = \log 3 / \log 2 \approx 1.585$$

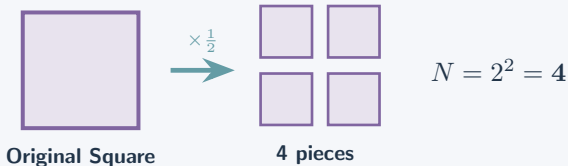
Self-Similarity & Scaling

Scaling a Line ($D = 1$): Scale by $r = \frac{1}{2} \implies 2$ pieces. Scale again by $\frac{1}{2} \implies 2 \times 2 = 4$ pieces.



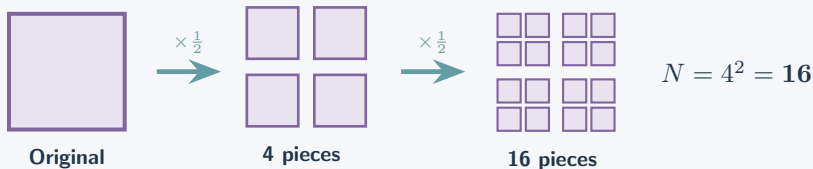
Self-Similarity & Scaling

Scaling a Square ($D = 2$) — Step 1: Scale by $r = \frac{1}{2} \implies$ **4 self-similar pieces** (2×2 grid).



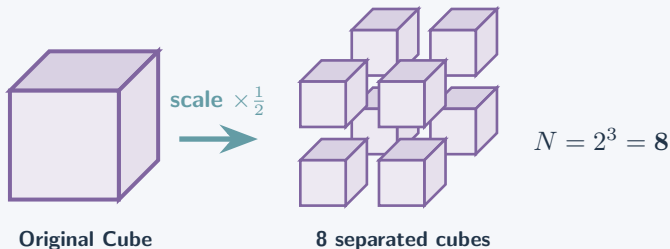
Self-Similarity & Scaling

Scaling a Square ($D = 2$) — Step 2: Scale each piece by $\frac{1}{2}$ again $\implies 4 \times 4 = \mathbf{16}$ self-similar pieces (4×4 grid)!



Self-Similarity & Scaling

Scaling a Cube ($D = 3$): Scale by $r = \frac{1}{2} \implies$
8 self-similar pieces ($2 \times 2 \times 2$ grid in 3D).



Self-Similarity & Scaling

Do You See a Pattern?

Each scaling step multiplies pieces by $(1/r)^D$!

Shape	Scale	1st Step	2nd Step	D
Line	$1/2$	$2 = 2^1$	$4 = 2^2$	1
Square	$1/2$	$4 = 2^2$	$16 = 2^4$	2
Cube	$1/2$	$8 = 2^3$	$(64 = 2^6)$	3

The Pattern

$$\text{Number of pieces} = \left(\frac{1}{r}\right)^D$$



$$D = ?$$

What if the shape is a **fractal**?

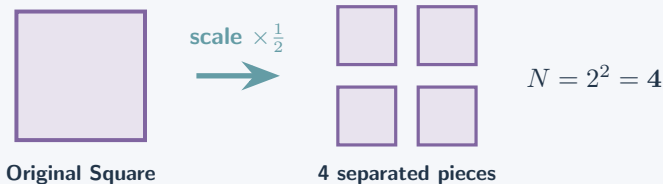
The Self-Similarity Rule

After scaling by factor r , the number of self-similar pieces is:

$$N = \left(\frac{1}{r} \right)^D$$

Square ($D = 2$) with scale factor $r = \frac{1}{2}$:

$$N = \left(\frac{1}{1/2} \right)^2 = 2^2 = 4 \text{ pieces}$$

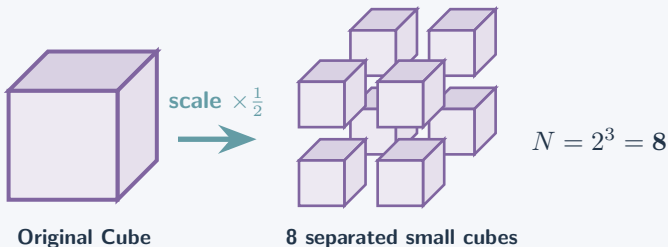


The Self-Similarity Rule

After scaling by factor r , the number of self-similar pieces is:

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Cube ($D = 3$) with scale factor $r = \frac{1}{2}$: $N = \left(\frac{1}{1/2} \right)^3 = 2^3 = 8$ pieces



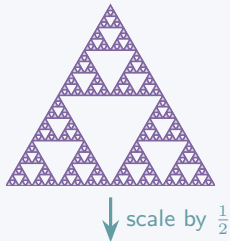
*Can we use this same rule to find the dimension of a **fractal**?*

Dimension of the Sierpinski Triangle

Let's apply the self-similarity rule to the **Sierpinski Triangle**.

Step 1

Scale the Sierpinski Triangle by $\frac{1}{2}$.

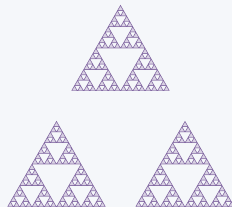


Dimension of the Sierpinski Triangle

Step 2 — Self-Similar Pieces

We get exactly **3** self-similar copies of the original triangle.

Each smaller triangle is a perfect replica, scaled by $r = \frac{1}{2}$.



3 copies

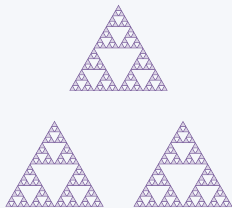
Dimension of the Sierpinski Triangle

Step 3 — The Equation

Using $N = \left(\frac{1}{r}\right)^D$:

$$3 = \left(\frac{1}{1/2}\right)^D$$

$$3 = 2^D$$



3 copies

Dimension of the Sierpinski Triangle

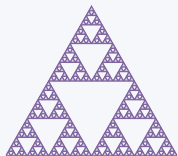
Step 4 — Solving for D

$$2^D = 3$$

$$D \log 2 = \log 3$$

$$D = \frac{\log 3}{\log 2}$$

$$D \approx \mathbf{1.585}$$



$$D \approx 1.585$$

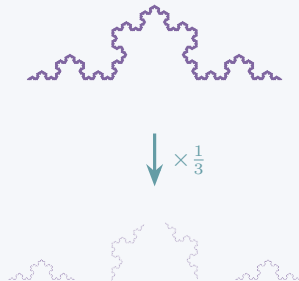
Not an integer! — strictly between 1D (line) and 2D (surface).

Similarly: The Koch Curve

The **Koch Curve** can be divided into **4** self-similar pieces, each scaled by $r = \frac{1}{3}$.

Setup

- Scale factor: $r = \frac{1}{3}$
- Self-similar pieces: $N = 4$



4 copies

Similarly: The Koch Curve

Koch Dimension

$$N = \left(\frac{1}{r}\right)^D$$

$$4 = 3^D$$

$$D = \frac{\log 4}{\log 3}$$

$$D \approx \mathbf{1.262}$$



$$D \approx 1.262$$

Also **non-integer** — between a line (1D) and a surface (2D), but closer to a line than the Sierpinski Triangle.

The General Theory: Box-Counting Dimension

The **Self-Similarity Dimension** works beautifully for self-similar fractals. But what about more complex shapes?

Box-Counting Dimension

The fractal dimension D is defined as:

$$D = \lim_{\varepsilon \rightarrow 0} \frac{\log N(\varepsilon)}{\log(1/\varepsilon)}$$

where $N(\varepsilon)$ is the number of boxes of side length ε needed to cover the fractal.

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Intuition

- Cover the fractal with a grid of boxes of size ε
- Count how many boxes $N(\varepsilon)$ the fractal touches
- Shrink $\varepsilon \rightarrow 0$ and observe how $N(\varepsilon)$ grows
- The **rate of growth** tells us the dimension

Handoff to Speaker 3

We've now uncovered a remarkable property of fractals:

- They have **non-integer dimensions** — genuinely “between” the familiar 1D, 2D, 3D.
- The Sierpinski Triangle lives in dimension $D \approx 1.585$.
- The Koch Curve lives in dimension $D \approx 1.262$.
- The **box-counting dimension** generalises this to *any* shape.

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We now know *what* fractals are and how to measure their complexity.

But **where do we see them?** And **how are they used** in computer science?

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We now know *what* fractals are and how to measure their complexity.

But **where do we see them?** And **how are they used** in computer science?

That's exactly where Speaker 3 takes us — fractals in nature, art, and computation.